

A Three-Branch Cross-Modal Interactive Network for RGB-D Salient Defect Detection

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Abstract—RGB defective images are abundant in color and texture, whereas depth images exhibit prominent defect shapes and boundaries. Given this fact, we propose a three-branch cross-modal interactive network for RGB-D salient defect detection (TCI-Net). Specifically, we first perform a real three-stream encoder-decoder network at both the image level and feature level, with each branch utilizing RGB, RGB-D, and depth images as input to fully extract the underlying complementary information from each modality. In the encoder stage, a differential guidance module (DGM) is proposed to guide the RGB branch in learning the boundary shape features of defects, while a fusion perception module (FPM) is devised to facilitate the depth branch in encoding more textual knowledge. In addition, we propose a cross-modal feature refinement module (CFRM) to bridge the feature gap between modalities and enhance information interaction. Finally, in the decoder stage, we incorporate boundary map supervision and a semantic guidance module (SGM) to enhance the details and contextual semantics of the defects, while gradually reconstructing the spatial scale. Extensive experiments and analyses on the RGB-D defect dataset NEU RSDDS-AUG demonstrate that the proposed TCI-Net significantly improves the segmentation accuracy compared with state-of-the-art algorithms.

Index Terms—Attention mechanism, feature refinement, multimodal, saliency detection, surface defect.

I. INTRODUCTION

SALIENT defect detection aims to identify the most visually attractive defective areas in images, which is crucial for ensuring product quality and safety, as well as for improving production efficiency. For example, defects on the surface of rails (such as holes and cracks), if not detected and repaired in time, may result in serious safety hazards. Traditional defect detection mainly relies on experienced inspectors for visual inspection, which requires significant human resources and is also susceptible to subjective factors. In recent years, with the rise of deep learning technology, automatic defect detection methods, especially those based on convolutional neural networks (CNNs), have been widely applied in surface defect detection. The accurate output of defect segmentation

Received 6 May 2025; revised 24 June 2025; accepted 8 July 2025. Date of publication 1 September 2025; date of current version 9 September 2025. This work was supported in part by China Postdoctoral Science Foundation under Grant 2021TQ0301; and in part by the National Natural Science Foundation of China under Grant 62106232, Grant 62102371, Grant 62172371, Grant 62036010, and Grant U21B2037. The Associate Editor coordinating the review process was Dr. Yi Qin. (Corresponding author: Mingliang Xu.)

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Digital Object Identifier 10.1109/TIM.2025.3599282

models can assist railway maintenance personnel in quickly locating and quantifying defect areas, thereby supporting timely decision-making and reducing potential operational risks.

Affected by complex industrial environments, such as lighting conditions, defects in RGB images often exhibit subtle shapes and boundaries, while depth images contain visually salient boundary representations. Therefore, effectively integrating RGB-D images can provide more comprehensive and accurate defect features for detection. Currently, mainstream detection algorithms based on CNN and RGB-D can be broadly categorized into three types: single-branch, dual-branch, and triple-branch networks.

As shown in Fig. 1(a), single-branch models [1], [2], [3] typically employ an early image-level fusion strategy by concatenating RGB and depth images into four channels as input to extract defect features. These algorithms have a simple, lightweight, and efficient model structure, but directly concatenating raw RGB and depth data may lead to feature space incompatibility due to modal differences. Dual-branch models [4], [5], [6], [7], as shown in Fig. 1(b), are currently the most widely used structures in RGB-D-based defect detection tasks. Such algorithms mainly process RGB and depth images through two independent streams and perform intermodality interaction through operations such as feature addition in the encoder or decoder stage. However, directly fusing RGB and depth features overlooks the issue of feature misalignment between different modalities. Feature-level triple-branch models [8], [9], [10] comprise RGB, depth, and feature fusion branches, as shown in Fig. 1(c). The feature fusion branch integrates cross-modal features to form an independent branch for encoding and decoding. Such algorithms greatly improve defect detection performance but lack cross-modal interaction at the image level, thus failing to fully encode the underlying structural knowledge of subtle defects and boundaries.

Based on the above considerations, we propose a novel three-branch cross-modal interaction network for RGB-D salient defect detection, as illustrated in Fig. 1(d). Specifically, to fully explore the low-level structural characteristics of subtle defects, we initially perform the image-level fusion of RGB and depth images using the Mixup [11] technique to generate RGB-D images, and then input multimodal images into the network for simultaneous feature extraction. In the encoder stage, we design a differential guidance module (DGM) and a fusion perception module (FPM), where the former guides the RGB branch in learning boundary features and other

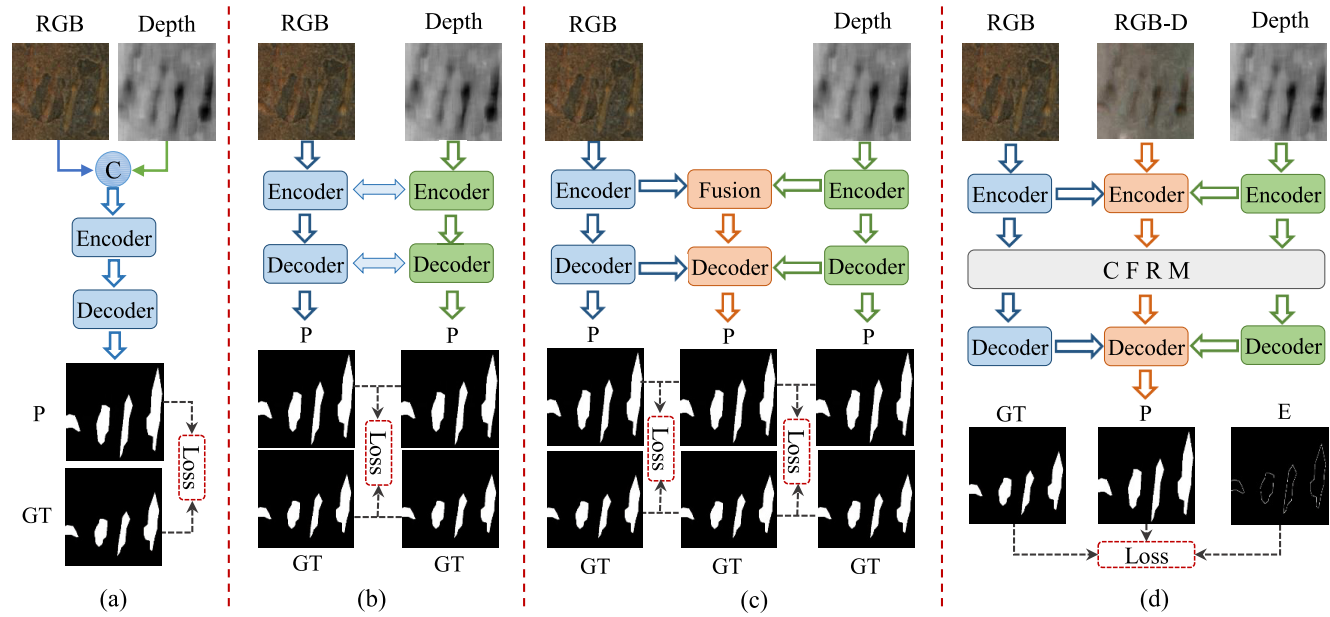


Fig. 1. Network architectures of RGB-D salient object detection. (a) Single branch. (b) Dual branch. (c) Feature-level three branches. (d) Image-level three branches (ours).

related information, while the latter directs the depth branch to assimilate color and texture information.

In addition, at the end of the encoder stage, we devise a cross-modal feature refinement module (CFRM) by applying spatial attention from the RGB branch and channel attention from the depth branch to the respective opposite branches, thereby further alleviating the feature misalignment between different modalities. Unlike conventional cross-modal feature fusion that applies a uniform fusion mechanism across all branches, the proposed refinement strategies are designed according to each modality's intrinsic properties, allowing dedicated feature extraction in each processing branch.

In the decoder stage, we incorporate supervised learning of boundary maps to enhance edge representations of subtle defects on low-resolution feature maps. We also develop a semantic guidance module (SGM) to improve the spatial contextual knowledge of high-resolution feature maps. Eventually, the network produces more accurate and comprehensive salient features for defect detection. In summary, the main contributions of this article are as follows.

- ① We innovatively propose a real three-branch network based on image-level and feature-level fusion to fully extract the underlying structural knowledge in RGB and depth images. In the encoder stage, the DGM is proposed to guide the RGB feature in learning more scene details and spatial relationships, while the FPM leads the depth feature in perceiving the semantics and colors of defects.
- ② We delicately propose a CFMR specifically for the RGB, depth, and RGB-D branches. CFMR adopts attention mechanisms and dilated convolutions and is tailored to alleviate the issue of feature misalignment across modalities and enhance multiscale feature representations.

- ③ We introduce edge map supervision and propose the SGM in the decoder stage to enhance the boundaries and contextual semantics of defects. The SGM employs a multistage and layer-by-layer guidance strategy to decode the salient spatial structures and semantic information. 采用多阶段、分层次式的引导策略来解码
- ④ We propose a three-branch cross-modal interactive network for RGB-D salient defect detection (TCI-Net). Extensive experiments and analyses conducted on the multimodal defect dataset NEU RSDDS-AUG demonstrate that TCI-Net significantly improves the salient defect detection performance compared with current SOTA algorithms.

II. RELATED WORKS

A. RGB Defect Detection

With the continuous advancement of deep learning, defect detection performance has significantly improved. Several works have focused on using RGB data to achieve accurate defect detection. Dong et al. [12] proposed a multisupervision framework that processes multiscale information for defect detection. However, the method relies solely on RGB data and struggles with complex backgrounds or defects with blurry edges. Song et al. [13] designed an encoder-decoder residual (EDR) network that achieves excellent results for surface defect detection, but lacks fine-grained structural modeling and may misidentify weak defects. Zhang et al. [14] developed an end-to-end multistage segmentation network for railway rail defect detection but does not utilize cross-modal fusion. Similarly, Zhou et al. [15] introduced an edge-aware multistage interaction (EMI) network with feature integration and edge optimization to improve the accuracy of surface defect detection. Han et al. [16] proposed a two-stage edge reuse network for steel surface defect detection, featuring both

prediction and refinement stages. Ding et al. [17] developed a framework (CSEP) that explores the relationships among features of varying scales. Zhou et al. [18] proposed an end-to-end attention-guided network (DAC) utilizing cascade processing for surface defect prediction.

Although these RGB-based methods have achieved notable improvements, they often rely on single or dual-stream networks for feature encoding. When dealing with complex scenes or situations involving subtle defects, issues such as false or missed detections are prone to occur. In contrast, TCI-Net fuses RGB and depth information throughout the feature processing pipeline, which not only enhances the network's ability to perceive structural boundaries and weak defects but also significantly reduces detection instability caused by modality discrepancies. This leads to greater robustness and adaptability in real-world industrial applications.

B. RGB-D Saliency Detection

Combining RGB and depth data has proven to enhance saliency detection by utilizing the complementary information provided by both modalities. Such algorithms can be roughly divided into single-branch, dual-branch, and three-branch networks.

For single-branch models, Zhao et al. [1] leveraged the power of pretrained networks with a single-stream encoder to propose a real time and robust salient object detection, but the lack of explicit modality separation limits fine-grained cross-modal differentiation. Zhang et al. [19] introduced the first uncertainty-inspired RGB-D saliency detection model based on a conditional variational autoencoder, improving robustness but lacking precision in structural detail. Ji et al. [2] proposed a collaborative learning framework that effectively integrates edge, depth, and saliency features. Wang et al. [20] developed a multimodal network (CLA) for railway defect detection using RGB-D images. Despite the success of these approaches, they often overlook the discrepancies between the RGB and depth modalities, which limits their ability to fully capture cross-modal interactions.

For two-branch models, Li et al. [4] introduced an attention-guided intertwined fusion network that captures cross-channel complementarity via progressive interaction, employing a deep supervision attention mechanism to measure salient areas. Li et al. [5] employed a late fusion strategy that combines high-level RGB and depth features from two independent branches at the decoder stage. Zhai et al. [6] designed a cascade refinement network that integrates multimodal and multilevel features, allowing RGB and depth modalities to enhance one another. Zhang et al. [7] explored the inconsistent interaction between RGB and depth modalities, proposing an interaction pattern in which the cross-modal interaction guides saliency detection. While these two-branch approaches have demonstrated the potential of fusing RGB and depth knowledge to improve detection performance, defects with faint features or complex backgrounds often lead to misalignment in spatial data across different modalities.

The feature-level three-branch models consist of RGB, depth, and feature fusion branches, where the feature fusion branch operates independently by integrating cross-modal

features for encoding and decoding. Cong et al. [10] introduced a cross-modal interaction and refinement network, which incorporates a cross-modal interaction mechanism that fully leverages the complementarity between RGB and depth features. Such algorithms greatly improve defect detection performance, but often lack cross-modal interaction at the image level, making it challenging to completely encode the underlying structural information of subtle defects and boundaries.

To address these issues, we propose a novel RGB-D salient defect detection method based on a three-branch cross-modal interaction framework. To our knowledge, the proposed TCI-Net is the first attempt to integrate both image-level and feature-level fusion mechanisms in an RGB-D industrial defect detection model. Image-level fusion not only enriches the model's perception of multiple attributes such as texture, edges, and depth contours, but also serves as a form of data augmentation and regularization to some extent, reducing overfitting and improving the model's robustness to noise and defects across different scenarios.

III. METHOD

In this section, we first provide a brief description of the overall network of the proposed TCI-Net. Subsequently, we detail the structures of the DGM and FPM in the encoder stage and subsequently expound on the CFRM. Finally, the techniques and SGM in the decoder stage are illustrated.

A. Overall Structure

As shown in Fig. 2, the overall architecture of the TCI-Net comprises three parallel branches, where the RGB-D image is generated by mixing the RGB and the depth image using the Mixup technique. Initially, the RGB, depth, and RGB-D images are input into the backbone network for feature encoding across different modalities. In the encoder stage, we design the DGM and FPM to guide the RGB and depth branch in effectively learning the complementary information of another modality from the RGB-D features. Afterward, a CFRM is tailored to reduce the feature differences and misalignment between multimodal data while enhancing modality interaction through attention mechanisms. Subsequently, in the early decoder stage, we introduce edge map supervision to enhance the boundaries of subtle defects; in the late decoder stage, an SGM is proposed to fully integrate the salient spatial structure and semantics. Finally, the model makes predictions on the output feature maps of the fusion branch to obtain the final segmentation results.

B. Encoder Stage

1) *Three-Branch Backbone*: Due to the low contrast and faint features of certain defects, fully leveraging the underlying details and structural information is crucial for effective detection. To this end, we first employ the Mixup [11] technique to perform pixel-level fusion of RGB and depth images, generating the RGB-D image that preserves and integrates the underlying structural knowledge from both modalities. Specifically, we perform linear interpolation between the paired

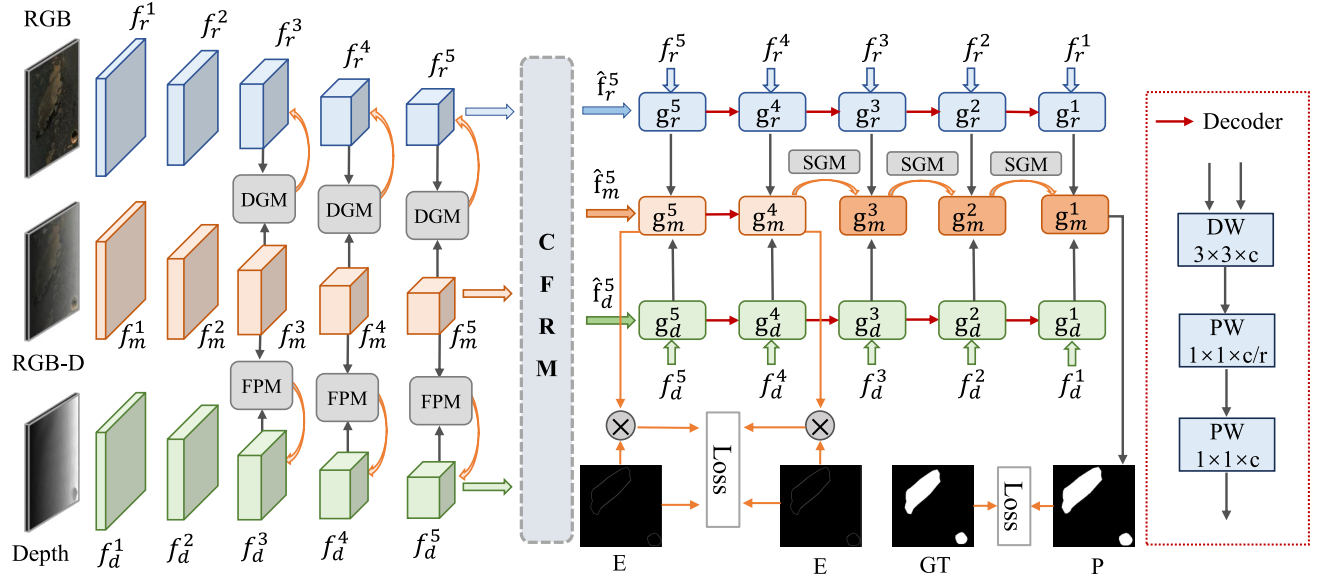


Fig. 2. Illustration of the overall architecture of the proposed TCI-Net, which consists of real three branches at the image level and feature level. GT and P indicate the ground truth and prediction saliency map. E refers to the edge map.

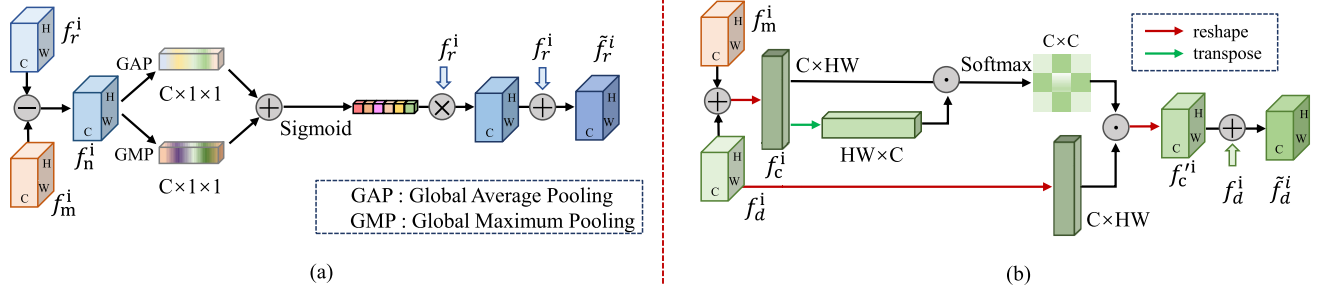


Fig. 3. Illustration of (a) DGM and (b) FPM.

RGB and depth images. Given an RGB image I_{rgb} and a corresponding depth image I_d , the fused RGB-D image I_{rgbd} is generated as

$$I_{\text{rgbd}} = \alpha \cdot I_{\text{rgb}} + (1 - \alpha) \cdot I_d \quad (1)$$

where $\alpha \in [0, 1]$ is a mixing coefficient. Based on experimental results, we set α to 0.5 to ensure equal contribution from both modalities, allowing the model to retain complementary structural and semantic information from RGB and depth sources. The Mixup technique not only enriches training data distribution but also serves as a regularization mechanism that encourages the model to behave linearly between training samples, thus improving generalization and robustness to subtle or noisy defects. The RGB image, depth image, and RGB-D image are then fed into a parallel three-branch network for feature extraction. Given the fact that depth, RGB, and RGB-D images are single-channel, three-channel, and four-channel images and become increasingly informative, respectively, we

employ different backbones for each branch, i.e., a shallow ResNet18 for the depth branch, a relatively deep ResNet34 for the RGB branch, and a deeper ResNet50 for the RGB-D

fusion branch. This configuration allows for optimal extraction of features relevant to each modality.

2) *Differential Guidance Module*: RGB images are rich in color and texture, but are sensitive to lighting conditions and lack boundary and depth information. Based on this, we propose the DGM to guide the RGB branch in learning more edge and depth signals, whose detailed structure is shown in Fig. 3(a).

Specifically, $f_r^i, f_m^i \in \mathbb{R}^{C \times H \times W}$ ($i = 3, 4, 5$) refer to the output feature maps of RGB branch and RGB-D branch of the i th layer, respectively. First, f_r^i and f_m^i are elementwise subtracted to obtain the differential feature map $f_n^i \in \mathbb{R}^{C \times H \times W}$. This operation is mainly to highlight the modality-specific discrepancies, i.e., emphasizing where RGB and RGB-D features diverge, which is particularly effective for detecting subtle structural differences. The f_n^i is passed through the global average pooling (GAP) and global maximum pooling (GMP) to obtain the channel weight matrices that are then added in an elementwise manner. The fused channel descriptor is passed through a Sigmoid activation function σ to produce a channelwise attention map. This attention map is then elementwise multiplied with f_r^i to perform adaptive feature

reweighting across channels, enhancing informative channels and suppressing irrelevant ones. Elementwise multiplication is commonly used for such attention modulation or gating mechanisms. Finally, a residual connection is applied by adding the original f_r^i back to the attention-enhanced feature to retain the original modality's information while benefiting from the attention guidance. This helps to stabilize training and improve gradient flow. The process can be described as

$$f_n^i = f_r^i \ominus f_m^i, \quad i = 3, 4, 5 \quad (2)$$

$$\tilde{f}_r^i = f_r^i \oplus (f_r^i \otimes \sigma(\text{GAP}(f_n^i) \oplus \text{GMP}(f_n^i))) \quad (3)$$

where $\ominus$, $\oplus$, and $\otimes$ denote elementwise subtraction, summation, and multiplication, respectively, and σ represents the Sigmoid activation function. The DGM highlights the structural features of defects via the weighted difference information between the fusion branch and the RGB branch, thus strengthening the boundary details and the depth features of the RGB branch.

3) *Fusion Perception Module*: Depth images exhibit prominent scene details and spatial relationships, while the color and texture features are relatively weak. On this basis, we design the FPM shown in Fig. 3(b) to guide the depth branch to perceive the semantics and colors of defects.

Specifically, let $f_m^i, f_d^i \in R^{C \times H \times W} (i = 3, 4, 5)$ be the output feature maps of the RGB-D branch and the depth branch at the i th layer, respectively. First, f_m^i and f_d^i are fused through elementwise summation. Then, we perform a reshape operation and generate $f_c^i \in R^{C \times HW}$ that is multiplied by the transposed feature map to produce the spatial attention map. Subsequently, the attention map is multiplied by the reshaped depth features to facilitate paying attention to the semantics and colors more efficiently, producing $f_c^{ri} \in R^{C \times H \times W}$ after restoring the feature scale. Finally, we perform a residual connection between f_c^{ri} and f_d^i and obtain $\tilde{f}_d^i$, which can be formulated as

$$f_c^i = R(f_m^i \oplus f_d^i), \quad i = 3, 4, 5 \quad (4)$$

$$f_c^{ri} = R(f_c^i) \odot \zeta(f_c^i \odot T(f_c^i)) \quad (5)$$

$$\tilde{f}_d^i = f_d^i \oplus f_c^{ri} \quad (6)$$

where R represents the feature reshape operation, ζ denotes the Softmax function, T signifies the transpose operation, and $\odot$ indicates the matrix multiplication. FPM enables the depth branch to learn rich semantics, colors, and texture from the fusion branch, providing a comprehensive and complementary feature representation for decoding.

C. Cross-Modal Feature Refinement Module

Despite including the DGM and FPM in the encoder stage, feature misalignment remains among the output feature maps of different modalities. Therefore, we develop the CFRM specifically for RGB, depth, and RGB-D branches to ensure more efficient and accurate decoding, the detailed structure of which is shown in Fig. 4.

As shown in Fig. 4(a), for the RGB branch, the spatial attention mechanism is first applied to the features $\tilde{f}_r^5$ output by the DGM to focus on the critical regions within images. Since RGB images are rich in semantic information and

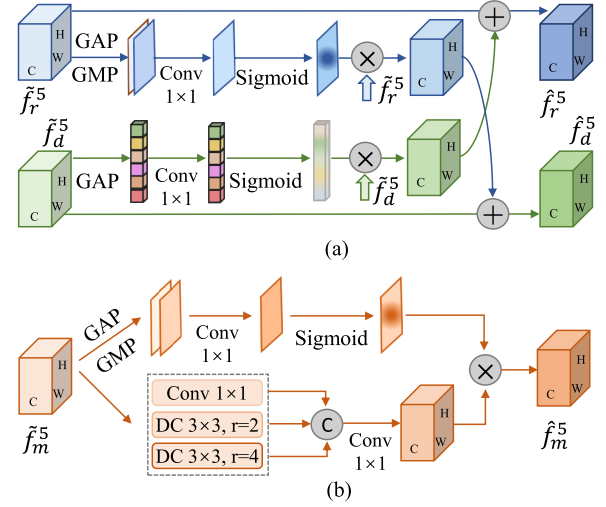


Fig. 4. Illustration of the CFRM of (a) RGB and depth branch and (b) RGB-D branch.

complex spatial structures, applying spatial attention helps to effectively highlight important regional features and enhance the representation of structural information. The attention map is then elementwise multiplied with $\tilde{f}_r^5$ to strengthen the structural representation, and the resulting enhanced features are elementwise added to the depth features $\tilde{f}_d^5$ to guide the depth branch in encoding more contextual and texture information, the final output is denoted as $\tilde{f}_d^5$.

On the other hand, for the depth branch, a channel attention mechanism is applied to the depth features $\tilde{f}_d^5$ output by the FPM to better exploit the importance of different channels. Since depth images exhibit prominent defect shapes and boundaries, applying channel attention helps to enhance the expressive capability of depth features. The generated attention map is elementwise multiplied with $\tilde{f}_d^5$ to selectively enhance informative channels. The attended features are then elementwise added to $\tilde{f}_r^5$ to guide the RGB branch in learning more enriched cross-channel boundary information, and the final output is denoted as $\tilde{f}_r^5$. The process can be represented as

$$\tilde{f}_d^5 = \tilde{f}_d^5 \oplus (\tilde{f}_r^5 \otimes \sigma(\text{Conv}(\text{Cat}(\text{GAP}(\tilde{f}_r^5), \text{GMP}(\tilde{f}_r^5)))) \quad (7)$$

$$\tilde{f}_r^5 = \tilde{f}_r^5 \oplus (\tilde{f}_d^5 \otimes \sigma(\text{Conv}(\text{GAP}(\tilde{f}_d^5)))) \quad (8)$$

where Conv refers to a 1×1 convolution and Cat denotes the operation of feature concatenation.

For the fusion branch, the output feature maps already contain rich and comprehensive multimodal information. To further extract the multiscale features of defects, we employ dilated convolutions with different rates to broaden the receptive field sizes of neurons. As shown in Fig. 4(b), for the output feature map $\tilde{f}_m^5$ of the fusion branch, we first apply GAP and GMP to enhance the feature representation capabilities for subtle defects. The spatial attention map $\tilde{f}_m^5$ is generated after a Sigmoid function is applied. Concurrently, features with multiscale information $\tilde{f}_m^5$ are obtained through a 1×1 convolution and two 3×3 dilated convolutions (with dilation rates of 2 and 4, respectively) in parallel. Finally, $\tilde{f}_m^5$ is

generated by multiplying $\tilde{f}_m^5$ and $\tilde{f}_m'^5$ in an elementwise fashion. The process can be represented as

$$\tilde{f}_m^5 = \sigma(\text{Conv}(\text{Cat}(\text{GAP}(\tilde{f}_m^5), \text{GMP}(\tilde{f}_m^5)))) \quad (9)$$

$$\tilde{f}_m'^5 = \text{Conv}(\text{Cat}(\text{Conv}(\tilde{f}_m^5), \text{DC}_{r=2}(\tilde{f}_m^5), \text{DC}_{r=4}(\tilde{f}_m^5))) \quad (10)$$

$$\hat{f}_m^5 = \tilde{f}_m^5 \otimes \tilde{f}_m'^5 \quad (11)$$

where DC stands for a 3×3 dilated convolution and r denotes the dilation rate.

CFRM exploits both spatial and channelwise relationships to align and recalibrate cross-modal features before they are passed to the decoder, enhancing semantic consistency and suppressing modality-specific noise. Therefore, CFRM could effectively alleviate the issue of feature misalignment across modalities, providing more precise and effective feature representations for the decoder stage. Especially, CFRM improves the network's defect detection capability in regions with low contrast or ambiguous structures.

D. Decoder Stage

In the decoder stage, the RGB features ($g_r^i, i = 1, 2, 3, 4, 5$) and depth features ($g_d^i, i = 1, 2, 3, 4, 5$) are generated using the classic U-Net [21] through the skip connections with the corresponding features in the encoder stage. For the fusion features, we introduce boundary map supervision and SGM to reconstruct the spatial scale progressively and enhance the feature representations of defects.

1) *Edge Map Supervision*: The low-resolution feature maps are rich in semantic information but suffer from a lack of clear defect boundaries. To this end, we introduce edge map supervision in the fusion branch. Specifically, g_r^i and g_d^i are first fused through concatenation and undergo depthwise separable convolution. Subsequently, a fine-grained boundary map E is generated by applying the Canny edge detector [22] to the ground-truth segmentation map. To integrate this boundary information into the fusion features, we perform elementwise multiplication between the edge map E and the fused feature map. This operation amplifies the response in boundary regions while suppressing irrelevant background noise, thereby guiding the model to focus on defect contours more precisely. Finally, we apply elementwise addition to further enhance the contextual richness and generate the boundary-aware fusion features $g_m^i (i = 4, 5)$. The process can be expressed as

$$g_m^5 = E \otimes (\tilde{f}_m^5 \oplus (\text{PW}_{\times 2}(\text{DW}(\text{Cat}(g_r^5, g_d^5)))))) \quad (12)$$

$$g_m^4 = E \otimes (\tilde{f}_m^4 \oplus (\text{PW}_{\times 2}(\text{DW}(\text{Cat}(g_r^4, g_d^4)))))) \quad (13)$$

where DW stands for the depthwise convolution and $\text{PW}_{\times 2}$ refers to the two sequential pointwise convolutions.

2) *Semantic Guidance Module*: For the relatively high-resolution feature maps, we propose the SGM to guide the subsequent layer in enhancing the high-level contextual information, as shown in Fig. 5. Let $g_m^i (i = 1, 2, 3)$ be the output feature map of the fusion branch at the i th layer. It first undergoes two 1×1 convolutions to obtain $g_m'^i$, and simultaneously, it goes through GAP followed by two 1×1 convolutions to get the channel attention map $g_m''^i$. After elementwise multiplication of g_m^i and $g_m''^i$, an upsampling

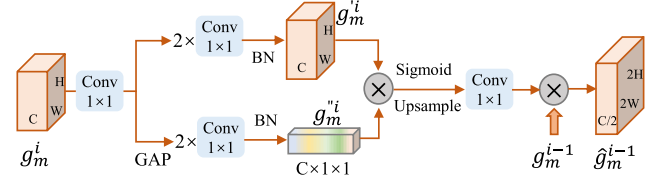


Fig. 5. Illustration of SGM.

operation is performed to adjust the scale of the feature map to be consistent with that of the $(i-1)$ th layer. Finally, the output features are multiplied by g_m^{i-1} , producing $\hat{g}_m^{i-1}$. The procedure can be expressed as

$$g_m^i = \text{PW}_{\times 2}(\text{DW}(\text{Cat}(g_r^i, g_d^i))), \quad i = 1, 2, 3 \quad (14)$$

$$g_m'^i = \text{BN}(\text{Conv}_{\times 3}(g_m^i)) \quad (15)$$

$$g_m''^i = \text{BN}(\text{Conv}_{\times 2}(\text{GAP}(\text{Conv}(g_m^i)))) \quad (16)$$

$$\hat{g}_m^{i-1} = g_m^{i-1} \otimes (\text{Conv}(U(\sigma(g_m'^i \otimes g_m''^i)))) \quad (17)$$

where BN refers to the batch normalization, U denotes the upsampling operation, and $\text{Conv}_{\times n}$ indicates the n consecutive 1×1 convolutions. The SGM employs a multilevel, multistage, and layer-by-layer guidance strategy to decode the salient spatial structure and semantic information. Finally, the salient map of defects is predicted on the last output feature map of the fusion branch.

E. Loss Function

During the training phase, the overall objective function is the sum of the boundary map prediction loss and the salient defect prediction loss. Specifically, the binary cross-entropy (BCE) loss measures the difference between the predicted probability distribution and the true probability distribution; the intersection over union (IoU) loss assesses the degree of overlap between the prediction and the object; and the Dice similarity coefficient (Dice) loss further emphasizes the contribution of the object area. The overall loss function can be formulated as

$$L_{\text{total}} = \sum_{i=4}^5 [\text{BCE}(S_i, G) + \text{IoU}(S_i, G)] + \text{Dice}(S, G) \quad (18)$$

where S_i and S represent the predicted edge map and salient map, respectively, and G signifies the ground-truth map.

IV. EXPERIMENTS

A. Experimental Setup

We conduct all experiments using PyTorch 1.10.0 on a computing platform equipped with a GeForce RTX 4090 graphics card (with 24 GB of video memory), an Ubuntu 20.04.6 LTS 64-bit operating system, and Python 3.7. During training, the image size is uniformly adjusted to 352×352 pixels as input. The model is trained using the Adam optimizer [31] with an initial learning rate of $1e^{-4}$. A cosine annealing learning rate strategy is employed with a weight decay of 0.0005. The training batch size is set to 16 with a total of 600 epochs of iteration. To ensure fairness, the training and testing configurations for other comparative algorithms are consistent with those used for TCI-Net presented in this article.

B. Datasets and Evaluation Metrics

1) *Datasets*: The experiments are conducted on the multi-modal defect dataset NEU RSDDS-AUG [20], which collects P60 rail images using binocular stereo vision between 130 and 400 mm. The defect locations in the images are randomly distributed to more realistically reflect the actual scene of railway track defect detection. NEU RSDDS-AUG presents several challenging characteristics, including subtle and faint defects that are easily overwhelmed by noise, complex backgrounds containing nondefect patterns such as rust and stains, significant lighting variations caused by different environmental conditions, and a wide range of defect scales. A total of 1862 images are produced after data augmentation and divided into training and testing sets in a ratio of 8:2, resulting in 1500 training images and 362 testing images.

2) *Evaluation Metrics*: To evaluate the performance of the proposed TCI-Net, we employ common salient object detection metrics, primarily including the precision–recall (P – R) curve, E -measure (E_m) [32], S -measure (S_m) [33], F -measure (F_m) [34], and mean absolute error (MAE) [35].

The P – R curve illustrates the precision and recall of a model at various thresholds, serving as an intuitive graphical tool for evaluating model performance. The larger the area under the curve, the better the model's performance. The formulas for calculating precision (P) and recall (R) are as follows:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN} \quad (19)$$

where TP, FP, and FN denote the true positives, false positives, and false negatives, respectively.

The E -measure is to evaluate the alignment between a binary foreground map and its corresponding ground truth by considering both global image statistics and local pixel matching information. The calculation formula for E -measure is given by

$$E_m = \frac{1}{H \times W} \sum_{y=1}^H \sum_{x=1}^W \phi(x, y) \quad (20)$$

where ϕ represents the matrix entry at pixel position (x, y) .

The S -measure represents the structural similarity between the predicted salient map and ground-truth map, which can be described as

$$S_m = \partial \times S_o + (1 - \partial) \times S_r \quad (21)$$

where ∂ is the weight coefficient, which is set to 0.5 here to balance the regional similarity S_r and the object similarity S_o .

The F -measure is a commonly used metric for evaluating classification performance, obtained by calculating the harmonic mean of precision and recall. The calculation formula is as follows:

$$F_m = \frac{(1 + \beta^2) \cdot (P \times R)}{\beta^2 \times (P + R)} \quad (22)$$

where β^2 is set to 0.2.

MAE score calculates the mean absolute difference between the predicted salient map S and the corresponding ground-truth map G , which can be expressed as

$$MAE = \frac{1}{H \times W} \sum_{y=1}^H \sum_{x=1}^W |S(x, y) - G(x, y)| \quad (23)$$

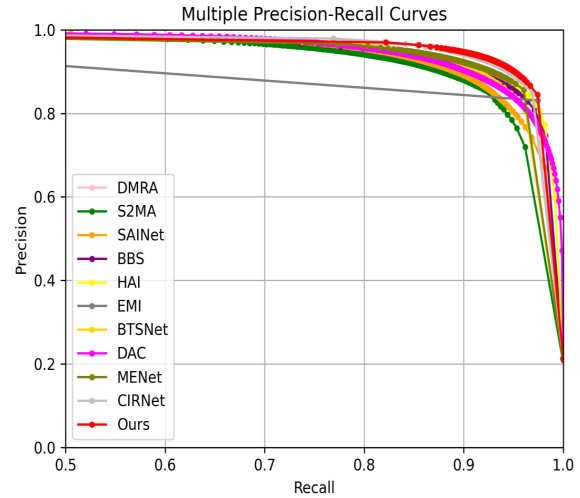


Fig. 6. Comparisons of P – R curve across different methods.

where H and W represent the height and width of the image, respectively.

C. Comparison of Experimental Results

To evaluate the effectiveness of the proposed method, we compare TCI-Net with several state-of-the-art object detectors, including both general object detectors and defect detectors. The comparative results, as presented in Table I, reveal that TCI-Net exhibits a significant advantage over other methods across most evaluation metrics. Specifically, TCI-Net achieves impressive performance with the S_m of 0.861, E_m of 0.937, F_m of 0.901, and MAE of 0.049, outperforming other advanced detection algorithms in these metrics. The MAE reveals that the mean absolute difference in the salient maps predicted by TCI-Net is smaller and more precise, which can be attributed to the effective cross-modal information fusion achieved by the proposed CFRM. In addition, the IoU, recall, and precision of TCI-Net rank first (79.25%), second (88.24% versus 91.67%), and third (87.22% versus 87.27%), respectively. The $F1$ score, which measures overall performance, reaches the highest (87.59%) for TCI-Net among all the detectors. In terms of inference speed, TCI-Net achieves 48.7 frames/s, which meets the real-time detection requirement while maintaining competitive accuracy.

Moreover, the P – R curve of different evaluation methods are illustrated in Fig. 6, showing that TCI-Net strikes a better balance between precision and recall than other algorithms. To more intuitively demonstrate the detection performance of TCI-Net, we compared the visualization results of different methods, as shown in Fig. 7. It can be observed that TCI-Net segments the overall defects more completely compared with other algorithms, as seen in the first row. Moreover, as clearly observed in the sixth row, the boundaries predicted by TCI-Net are smoother and more continuous. Especially in low-contrast or complex-texture regions, the edge details are better preserved, further enhancing the accuracy of defect localization. The results validate that boundary supervision

TABLE I
COMPARISONS OF DETECTION RESULTS ON NEU RSDDS-AUG

Method	$S_m \uparrow$	$E_m \uparrow$	$F_m \uparrow$	MAE $\downarrow$	Precision	Recall	F1	IoU	FPS
<i>General Detectors</i>									
DMRA [23]	0.736	0.834	0.783	0.141	80.36%	74.18%	77.07%	62.96%	17.2
S2MA [24]	0.775	0.864	0.817	0.141	76.91%	82.83%	79.68%	70.39%	8.9
CIRNet [10]	0.801	0.894	0.856	0.086	87.26%	86.61%	86.93%	77.87%	60.4
BBS [6]	0.828	0.909	0.867	0.074	86.28%	87.35%	86.81%	77.31%	48.9
HAI [25]	0.718	0.829	0.803	0.171	73.90%	91.67%	81.94%	68.91%	33.6
<i>Defect Detectors</i>									
EMI [15]	0.800	0.876	0.850	0.104	82.65%	87.46%	85.06%	74.82%	56.3
BTSNet [26]	0.845	0.913	0.892	0.067	-	-	-	-	-
CSEP [17]	0.814	0.899	0.866	0.085	85.29%	87.61%	86.40%	76.65%	56.1
DAC [18]	0.824	0.920	0.878	0.069	86.71%	88.09%	87.23%	77.77%	47.5
SPNet [27]	0.842	0.920	0.889	0.064	84.34%	87.21%	85.74%	77.68%	41.0
RFNet [28]	0.842	0.923	0.899	0.063	-	-	-	-	-
MENet [29]	0.852	0.929	0.899	0.057	87.27%	86.19%	86.65%	77.93%	61.2
SAINet [30]	0.851	0.927	0.898	0.055	87.06%	86.34%	86.69%	78.12%	51.6
TCI-Net(Ours)	0.861	0.937	0.901	0.049	87.22%	88.24%	87.59%	79.25%	48.7

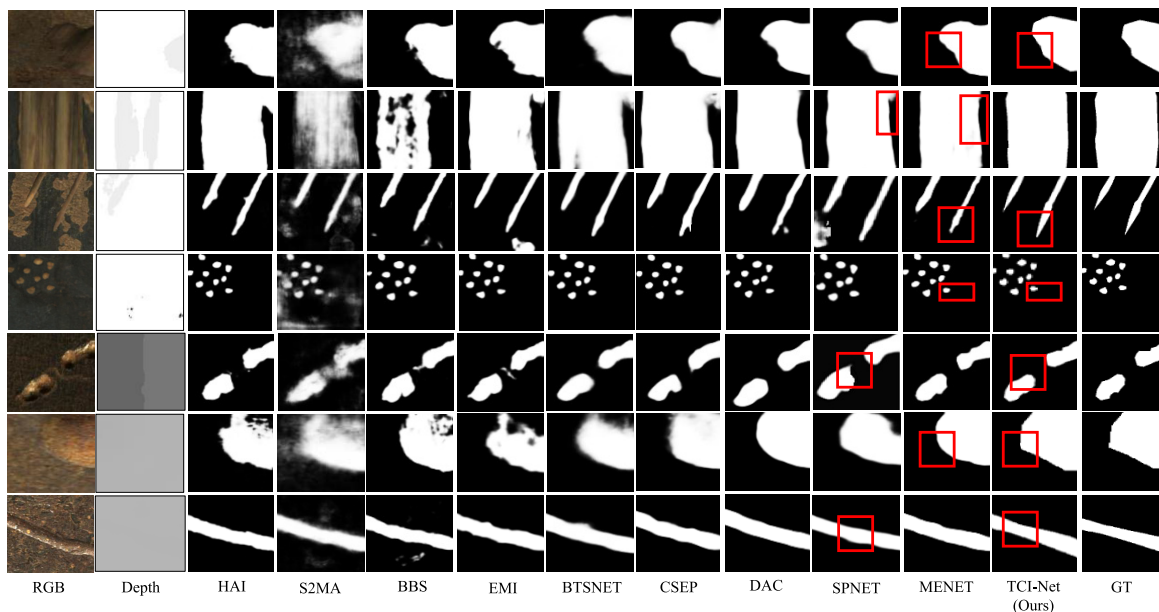


Fig. 7. Comparisons of visualization results across different methods.

and SGM in the decoder stage can significantly enhance the model's encoding capability for weak boundaries and defects.

D. Ablation Study

To better understand the structure and each module in TCI-Net, we conduct a series of ablation experiments on NEU RSDDS-AUG. We, respectively, verify the roles of the RGB-D image, DGM, FPM, CFRM, SGM, and the kernel sizes in the feature refinement module of the fusion branch.

1) *RGB-D Image, DGM, and FPM*: We conduct multiple sets of comparative experiments to verify the impact of image-level RGB-D fusion, DGM, and FPM, as shown in Table II. It can be observed that as we gradually introduce the RGB-D image, DGM, and FPM into the network, all the detection

TABLE II
ABLATION STUDIES ON RGB-D IMAGE, DGM, AND FPM

Method	MAE $\downarrow$	$S_m \uparrow$	$E_m \uparrow$	$F_m \uparrow$
Baseline	0.076	0.695	0.731	0.801
+RGB-D	0.072	0.765	0.810	0.832
+DGM	0.069	0.802	0.821	0.852
+FPM	0.065	0.901	0.846	0.861

metrics can be consistently improved compared with the baseline. The introduction of the image-level RGB-D fusion map provides more comprehensive information on the underlying detailed structure of defects. In addition, with the inclusion of DGM, the MAE is improved from 0.072 to 0.069, indicating

TABLE III
ABLATION STUDIES ON CFRM IN RGB, DEPTH, AND RGB-D BRANCHES

RGB	Depth	RGB-D	MAE↓
—	—	—	0.065
✓	—	—	0.063
—	✓	—	0.063
—	—	✓	0.062
✓	✓	—	0.060
✓	—	✓	0.059
—	✓	✓	0.058
✓	✓	✓	0.056

TABLE IV
ABLATION STUDIES ON DIFFERENT KERNEL SIZES IN CFRM

Kernel Size	MAE↓	S_m ↑	E_m ↑	F_m ↑
1×1, 1×1, 1×1	0.051	0.839	0.927	0.885
1×1, 3×3 ($r=2$), 3×3 ($r=2$)	0.055	0.840	0.862	0.863
1×1, 3×3 ($r=3$), 3×3 ($r=3$)	0.054	0.847	0.854	0.872
1×1, 3×3 ($r=4$), 3×3 ($r=4$)	0.053	0.842	0.868	0.887
1×1, 3×3 ($r=2$), 3×3 ($r=3$)	0.053	0.832	0.857	0.816
1×1, 3×3 ($r=3$), 3×3 ($r=4$)	0.056	0.843	0.841	0.845
1×1, 3×3 ($r=2$), 3×3 ($r=4$)	0.049	0.861	0.837	0.901

that the fusion branch guides the RGB branch to focus on more defect boundaries. Furthermore, the addition of FPM further improves the MAE to 0.065, suggesting that FPM provides the depth branch with rich information, such as texture and color, for accurate salient defect detection.

2) *Cross-Modal Feature Refinement Module*: To verify the impact of the CFRM on TCI-Net, we conduct ablation experiments on different modal refinement branches using the NEU RSDDS-AUG dataset. The experimental results are shown in Table III. The introduction of the refinement module to the RGB branch, depth branch, and fusion branch effectively improves the MAE of the baseline model, indicating that adopting the CFRM can enhance the complementary information between modalities while reducing the semantic and informational differences of cross-modal features. When combining these modules in pairs, the MAE is further improved and reaches its optimal value (0.056) when all three are integrated.

3) *Kernel Size in CFRM*: We conduct ablation experiments to investigate the impact of different kernel sizes and dilation rates in the fusion branch of the CFRM module, as shown in Table IV. When all three convolution kernels are set to 1×1, the model achieves relatively weak performance across all four metrics, indicating limited contextual awareness due to the small receptive field. Introducing dilated convolutions in the latter two branches [e.g., 3×3 ($r=2$) and 3×3 ($r=2$)] results in a slight performance improvement in S_m and F_m , but a noticeable drop in E_m , suggesting the model captures more structure but less precise edge details. Further increasing the dilation rate r to 3 yields better S_m (0.847) and F_m (0.872), indicating an enhanced ability to integrate contextual cues. However, when both dilation rates are set

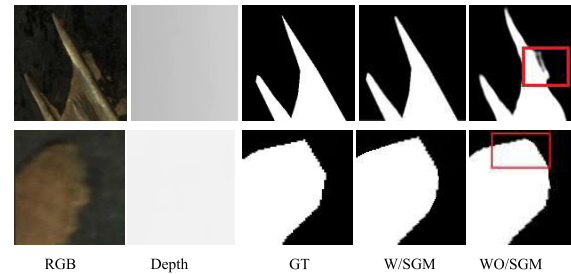


Fig. 8. Visualization results W/O SGM.

TABLE V
ABLATION STUDIES ON W/O THE SGM

Method	MAE↓	S_m ↑	E_m ↑	F_m ↑
WO/SGM	0.056	0.834	0.896	0.892
W/SGM	0.049(+0.5%)	0.861(+2.7%)	0.937(+4.1%)	0.901(+0.9%)

TABLE VI
ABLATION STUDIES ON W/O THE EDGE

Method	MAE↓	S_m ↑	E_m ↑	F_m ↑
WO/Edge	0.054	0.847	0.899	0.893
W/Edge	0.049	0.861	0.937	0.901

to 4, although F_m improves to 0.887, a degradation in E_m and S_m is observed, possibly due to over-enlarged receptive fields introducing irrelevant background noise. A combination of uneven dilation rates, such as (2, 3) and (3, 4), leads to unstable performance and fails to balance contextual richness and localization precision. The best overall results are achieved with the configuration 1×1, 3×3 ($r=2$), and 3×3 ($r=4$), which delivers the lowest MAE (0.049) and the highest F_m (0.901), demonstrating that this combination effectively balances local detail and global context. This suggests that moderate dilation diversity facilitates complementary feature extraction without overwhelming the network with noise. Therefore, we adopt 3×3 convolutions with dilation rates (2, 4) as the optimal setting in the fusion branch of CFRM.

4) *Semantic Guidance Module*: Due to a lack of semantics in high-resolution feature maps, we introduce the SGM through layer-by-layer guidance to decode both the salient spatial structure and high-level context. As shown in Table V, S_m , E_m , and F_m all have varying degrees of improvement when integrating SGM. Notably, the MAE significantly improves from 0.056 to 0.049 as well. Although the numerical improvement may appear limited, it holds significant practical value due to the sensitivity of MAE to overall image errors, particularly in boundary and detail regions. The visualization detection results are shown in Fig. 8. It can be observed that using SGM enables a more precise and complete segmentation of defect areas and boundaries.

5) *Edge Map Supervision*: To verify the effectiveness of the boundary supervision mechanism in improving defect detection performance, we conduct ablation experiments under

TABLE VII
PERFORMANCE COMPARISON ON PUBLIC SALIENCY DETECTION DATASET

Model	Type	STEREO [34]			NLPR [36]			NJUD [37]			LFSD [38]		
		$F_m \uparrow$	$S_m \uparrow$	MAE $\downarrow$	$F_m \uparrow$	$S_m \uparrow$	MAE $\downarrow$	$F_m \uparrow$	$S_m \uparrow$	MAE $\downarrow$	$F_m \uparrow$	$S_m \uparrow$	MAE $\downarrow$
DMRA [23]	Two	.886	.885	.047	.874	.889	.033	.888	.880	.052	.852	.839	.083
FRDT [39]	Two	.898	.900	.042	.897	.912	.029	.898	.899	.046	.855	.849	.080
SSF [40]	Two	.890	.892	.044	.898	.914	.025	.900	.900	.042	.862	.849	.075
A2dele [41]	Two	.886	.886	.043	.881	.897	.028	.873	.870	.051	.828	.825	.083
JL-DCF [3]	Single	.874	.885	.050	.891	.909	.029	.904	.902	.041	.821	.817	.103
PGAR [42]	Two	.900	.906	.042	.915	.929	.024	.906	.908	.042	.839	.844	.081
S2MA [24]	Two	.815	.842	.074	.901	.915	.029	.888	.894	.053	.831	.829	.101
DANet [1]	Single	.819	.841	.071	.902	.913	.028	.892	.897	.046	.841	.837	.103
BiANet [43]	Two	.884	.888	.049	.876	.900	.032	.912	.911	.039	.728	.742	.134
UCNet [19]	Two	.835	.824	.065	.903	.919	.025	.886	.897	.043	.859	.856	.073
BBS [6]	Two	.877	.898	.042	.915	.930	.023	.919	.920	.035	.868	.869	.072
ASIFNet [4]	Two	.880	.882	.048	.890	.907	.029	.888	.890	.047	.860	.852	.080
TCI-Net(Ours)	Three	.914	.916	.040	.925	.934	.025	.928	.918	.037	.884	.867	.069

two settings: with boundary supervision (W/Edge) and without boundary supervision (WO/Edge). The results are shown in Table VI. It can be observed that after introducing boundary supervision, all evaluation metrics exhibit noticeable improvements. Specifically, the MAE decreases from 0.054 to 0.049, indicating a further reduction in the error between the predicted results and the ground truth. Meanwhile, S_m improves from 0.847 to 0.861, E_m significantly increases from 0.899 to 0.937, and F_m rises from 0.893 to 0.901. These results demonstrate that boundary supervision can effectively guide the model to learn more accurate edge information, thereby enhancing the overall segmentation accuracy and contour completeness.

E. Experiments on Public Saliency Detection Datasets

To further validate the performance of TCI-Net, we conduct extensive experiments on four additional public RGB-D datasets. These datasets offer larger scales and greater diversity in terms of scene complexity, object appearance, and background variations, providing a more comprehensive evaluation of the model's generalization. The experimental results are shown in Table VII. On the STEREO dataset, TCI-Net achieves an MAE of 0.040, outperforming ASIFNet (0.048) by 0.008. On the NLPR dataset, TCI-Net achieves an F_m score of 0.925, which is 3.5% higher than ASIFNet (0.890). On the NJUD dataset, TCI-Net delivers the highest F_m of 0.928, along with competitive S_m (0.918) and MAE (0.037), showing strong performance in both boundary and object structure perception. For the LFSD dataset, which is particularly challenging due to weak boundaries and small-scale defects, TCI-Net achieves the best F_m of 0.884 and MAE of 0.069 as well. The results fully demonstrate the generalization capability and robustness of TCI-Net in salient object detection tasks across diverse datasets.

V. CONCLUSION

We propose a novel three-branch cross-modal interactive network for RGB-D salient defect detection, called TCI-Net. In the encoder stage, the image-level RGB-D fusion, DGM,

and FPM are introduced to capture the structural details of defects and ensure sufficient interaction between different modalities. Meanwhile, the CFRM is designed to mitigate feature discrepancies between cross-modal maps. In the decoder stage, we introduce edge map supervision and devise the SGM to progressively refine defect positions and boundaries. TCI-Net demonstrates superior detection performance on the NEU RSDDS-AUG industrial RGB-D dataset. Moreover, experiments on other saliency detection datasets also validate the generalization and robustness of TCI-Net. The accurate and efficient output of TCI-Net can assist railway maintenance personnel in early detection and localization of surface defects, thereby reducing operational risks and improving maintenance efficiency. Future work will focus on streamlining the network structure and reducing model complexity while maintaining detection accuracy for further optimization.

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